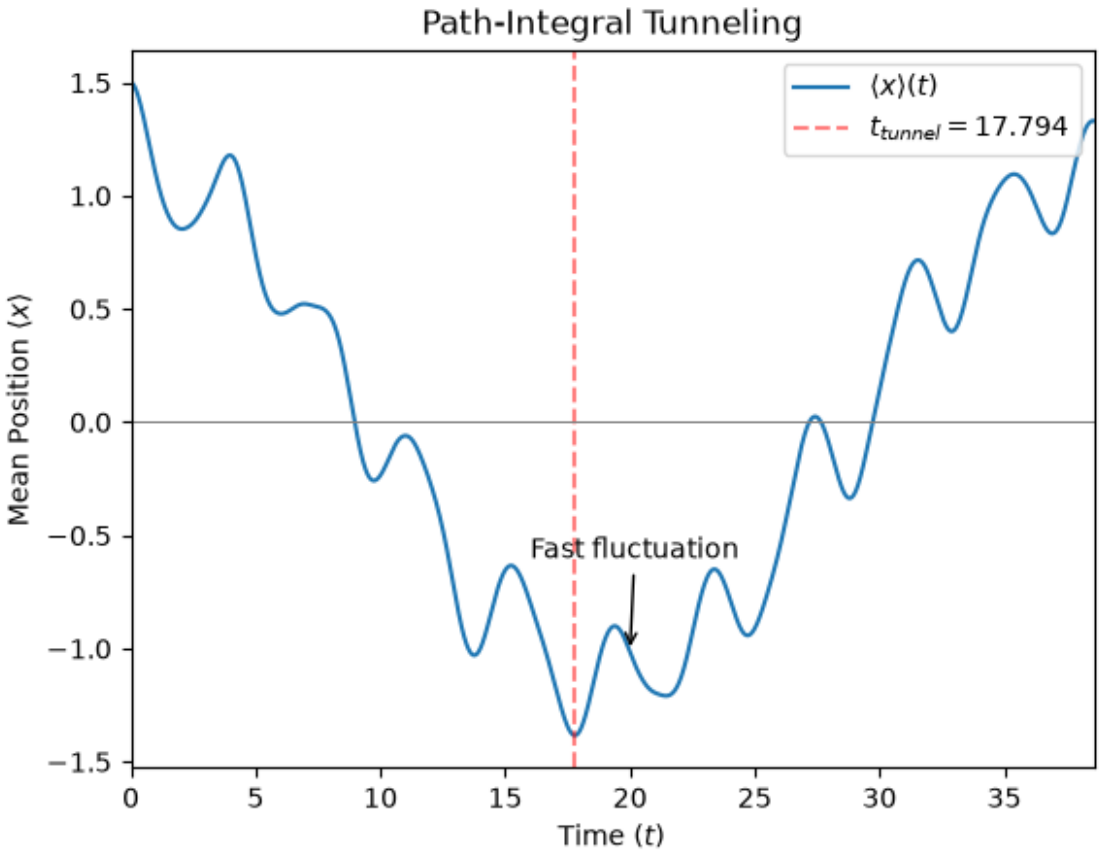
Assignment 2
PHYS 142
Aman Palanati
TSN: 200101723
$\alpha = 0.3 + \frac{723}{5000} = 0.4446$

Problem A:

(i) Path-Integral Evolution Plot



(ii) Interpretation

The first major minimum occurs at $t_{tunnel} = 17.7942$, with $\langle x \rangle = -1.3836$. The slow transfer shows that the mean position initially decreases as the particle tunnels from the right well to the left well, with t_{tunnel} indicating the particle's greatest mean displacement toward the left well. The mean position subsequently returns toward the right well, demonstrating tunneling between the two wells in both directions. There are also faster fluctuations superimposed on the slower tunneling oscillation. This behavior is not exhibited in the smooth sinusoidal two-state evolution in Problem D, which suggests that contributions from higher-energy eigenstates exist, since the initial wavefunction is not a superposition of only the two lowest energy eigenstates ϕ_0 and ϕ_1.

Problem B:

(i) Validation

Computed Eigenvalues

α	E_0	E_1
0.4	1.23447724	1.34690427
0.6	1.06247981	1.42104358

The computed eigenvalues agree with the published values to 4 decimal places with small differences in the order of 10^{-5}. Disagreement in further digits is likely because the grid was discretized using a finite Δx.

(ii) Convergence Analysis

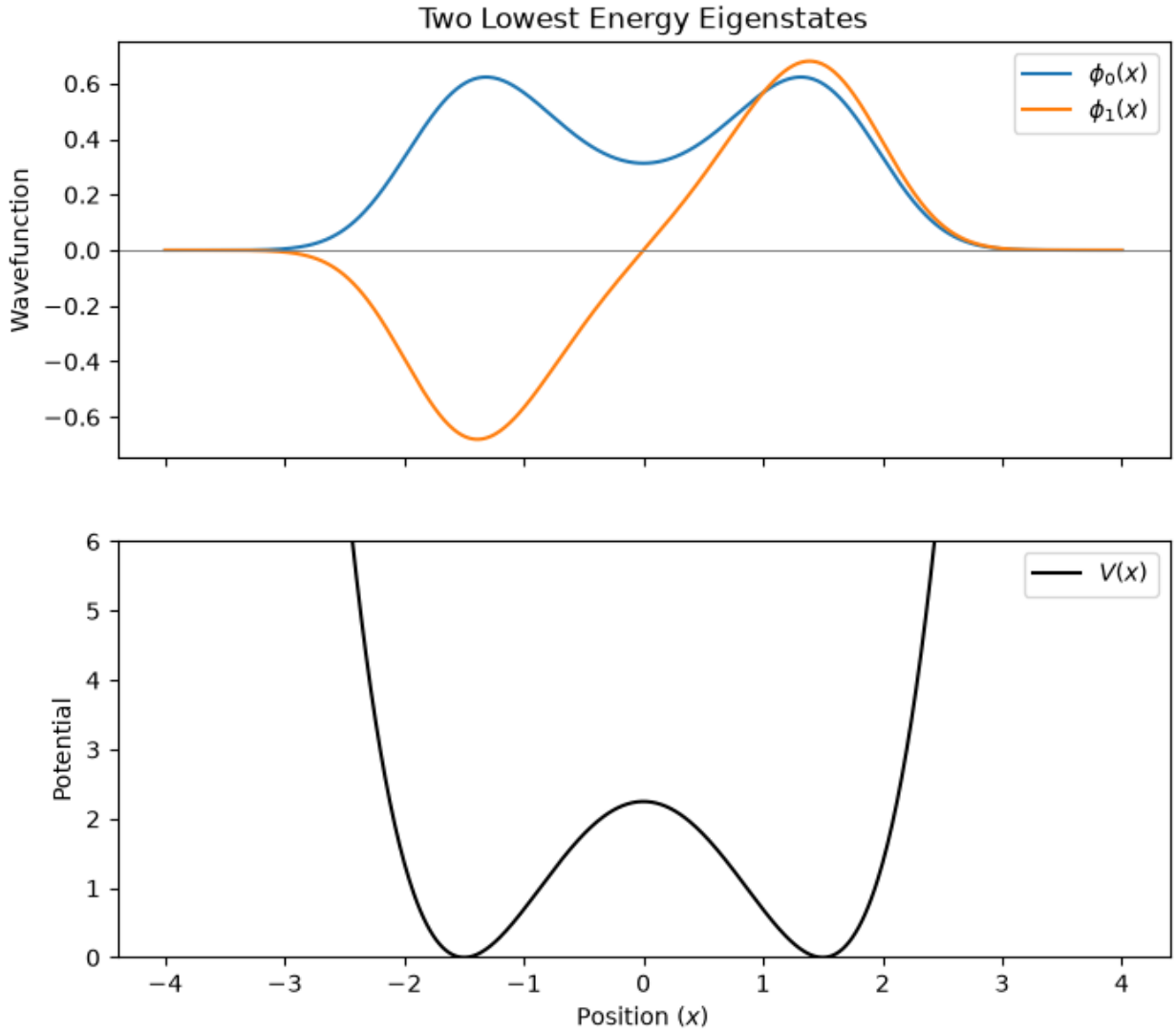
The original grid produced $E_0 = 1.19457165$, $E_1 = 1.35741495$, and $\Delta E = 0.16284330$.

Grid	E_0	E_1	ΔE	**Change in ΔE**
Original	1.19457165	1.35741495	0.16284330	0.00000000
Half spacing	1.19459193	1.35744304	0.16285111	0.00000781
Larger box	1.19457165	1.35741495	0.16284330	0.00000000

Halving the grid spacing increased the energy gap by 7.81×10^{-6}, a difference of approximately 0.0048%. In contrast, increasing the box size resulted in a negligible change. This indicates that the results are more sensitive to the grid spacing rather than the box size. The energy gap is reliable to approximately four digits after the decimal point ($\Delta E \approx 0.1628$), while additional digits are likely affected by error from discretization.

The code that sets the grid spacing is `dx_test = 2 * half_box / nd`, and the Hamiltonian is updated using `build_hamiltonian(x_test, dx_test, ALPHA)` within `solve_eigenstates()`.

(iii) Eigenfunction Plot



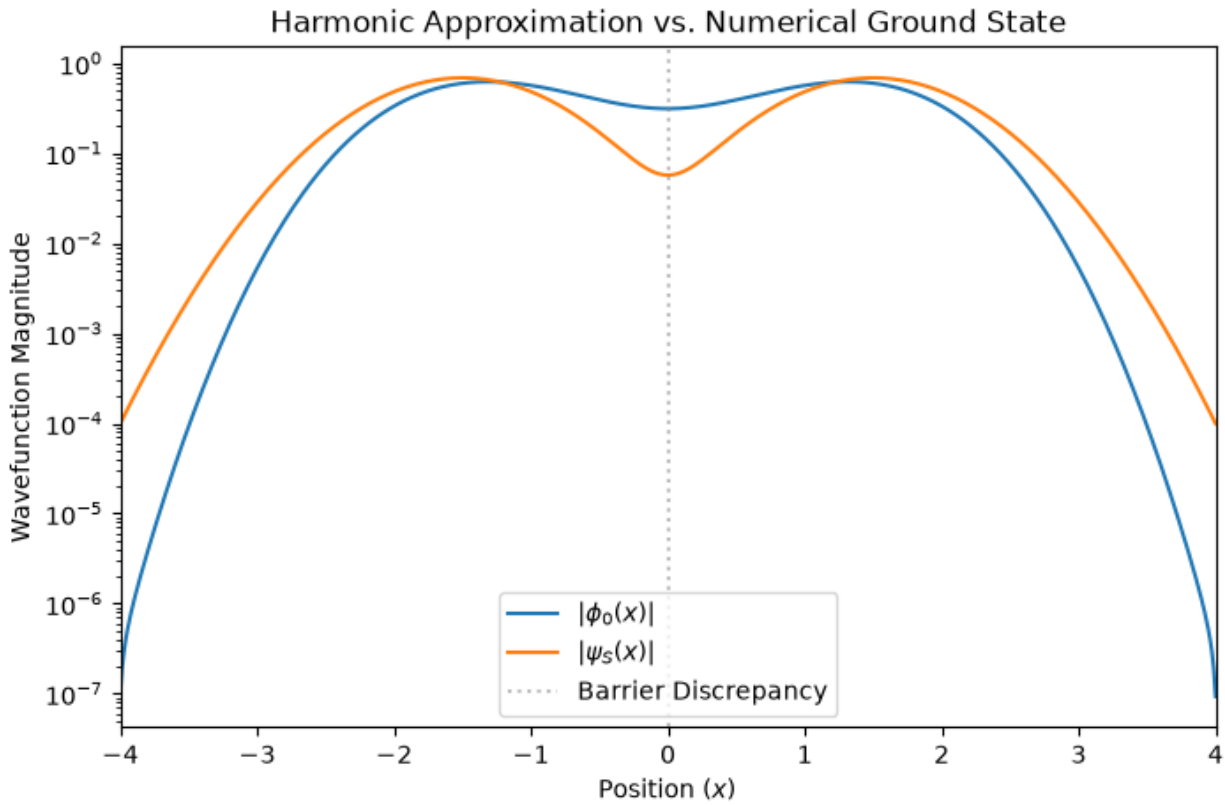
Problem C:

(i) Energy Comparison

Quantity	Numerical	Harmonic	Relative Error
Ground State Energy	1.19457165	1.44239200	20.75%
First Excited State Energy	1.35741495	1.46935648	8.25%
Energy Gap	0.16284330	0.02696448	83.44%

The individual energies are considerably more accurate in relative terms than their gap. The ground-state energy gap has a larger signed error (+0.24782035) than the first excited-state (+0.11194153). Subtracting these errors gives a signed gap error of -0.13587882, which explains why the harmonic approximation substantially underestimates the energy gap, as seen in the plot in part (ii).

(ii) Wavefunction Comparison



The wavefunctions agree reasonably well near the potential minima ($x = \pm x_{min}$). However, the harmonic approximation wavefunction magnitude is significantly smaller in the barrier region because the quadratic potential approximations used to create the harmonic wavefunctions do not accurately describe the full double-well potential away from the minima, which causes the discrepancy seen in the plot.
Since the energy expectation values depend on the wavefunctions, this discrepancy contributes to errors in the approximate energies. However, the energy gap's magnitude is much smaller than that of the energies, the relative error of the energy gap is more sensitive to the error introduced by the discrepancy, which is why it is so large compared to the relative errors in the energies.

Problem D:

(i) Analytical Derivation

$$\langle x \rangle(t) = \int \psi^*(x,t)\ x\ \psi(x,t)$$

The given two-state wavefunction equation uses normalized, orthogonal states, ensuring $\int |\psi|^2 dx = 1$. Using this equation and expanding gives:

$$\langle x \rangle(t) = \tfrac{1}{2}\left[\int x\phi_0^2 dx + \int x\phi_1^2 dx + e^{-i\Delta Et/\hbar}\int x\ \phi_0\phi_1 dx + e^{i\Delta Et/\hbar}\int x\ \phi_1\phi_0 dx\right]$$

Since ϕ_0 is even and ϕ_1 is odd, the first two integrals vanish due to symmetry:

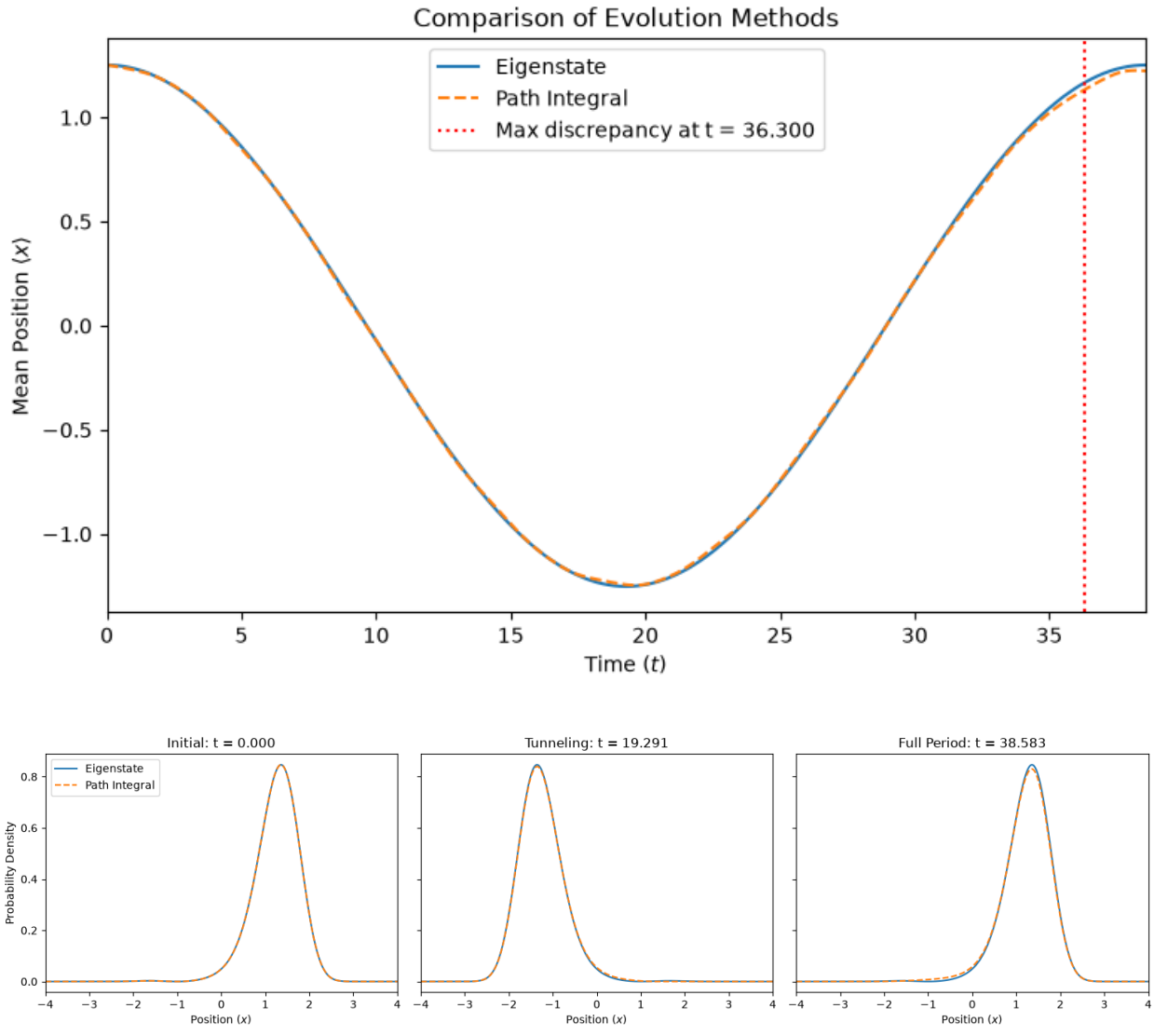
$$\langle x \rangle(t) = \tfrac{x_{01}}{2}\left(e^{-i\Delta Et/\hbar} + e^{i\Delta Et/\hbar}\right) = x_{01}\cos(\Delta Et/\hbar)$$

t_{tunnel} occurs at the first minimum, which is when $\Delta Et/\hbar = \pi$ so $t_{tunnel} = \frac{\pi\hbar}{\Delta E}$.

(ii) Path-Integral Evolution

Using the same initial state $\psi(x,0) = (\phi_0 + \phi_1)/\sqrt{2})$, I reused the path-integral propagator from Problem A to evolve the wavefunction for one full oscillation. The mean position was then calculated at each step.

(iii) Comparison of Methods



$$\max_{0 \le n\epsilon \le T} |\langle x \rangle_{\mathrm{PI}}(n\epsilon) - \langle x \rangle_{\mathrm{eig}}(n\epsilon)| = 0.03366176 \text{ at } t = 36.30012136$$
$$\text{with } \langle x \rangle_{\mathrm{PI}} = 1.13267130 \text{ and } \langle x \rangle_{\mathrm{eig}} = 1.16633306$$

The two methods agree initially at $t = 0$, but small discrepancies develop over time, becoming more noticeable toward the end of the oscillation. The largest discrepancy occurs at $t = 36.30012136$, although both methods continue exhibit the same tunneling behavior at all timesteps. One possible cause of this discrepancy buildup is the finite time step used to evaluate the path-integral propagator, which may accumulate phase errors during repeated propagation. A targeted check would

be to halve Δt, rebuild the propagator, and repeat the evolution. If time-step discretization is the dominant source of error, the path-integral result should be closer to the eigenstate prediction, reducing the maximum discrepancy.

AI Use Appendix:

1. **Tools and Usage:** ChatGPT was used for writing/debugging Python code, obtaining conceptual explanations for each of the problems, interpreting numerical results, structuring the report, and verifying submission requirements.

2. **Prompts:** My prompts focused on implementations, conceptual explanations, and troubleshooting: defining the harmonic approximation wavefunctions, constructing the path-integral propagator and measuring tunneling time in Problem A, calculating and normalizing numerical eigenstates using `eigsh` in Problem B, evaluating harmonic-approximation energies and interpreting wavefunction discrepancies in Problem C, deriving the two-state mean-position equation and comparing evolution methods in Problem D, and generating the `results.json` file.
   Full chat transcript: https://chatgpt.com/share/6aebc171-0448-83e8-b6a4-d7b9d207eb93.

3. **Errors and Verification:**

   - *Energy Gap Interpretation:* In Problem C, the model initially said the large relative error in the harmonic energy gap was due to inaccurate tunneling behavior. I questioned whether this directly explained the error and gave my reasoning that the unequal signed errors in the approximate energies and the small magnitude of the gap was a more accurate explanation.
   - *Harmonic Approximation:* The model explained the wavefunction discrepancy using the approximate potentials V_+ and V_- without initially clarifying their role in the calculation. I pointed out that these functions were not directly used in Problem C, which helped me clarify that they were used to derive the approximate wavefunctions.
   - *JSON Variable and Simulation Duration Errors:* The model's JSON export guidance referenced an inconsistent variable name (`x_mean` instead of `x_mean_pi`) and suggested directly using a simulation duration of $2\pi/\Delta E$, which was insufficient for the required JSON export times. This caused an `IndexError` when extracting values beyond the 1574 available time steps. I identified the issue by inspecting the array dimensions and corrected it by extending the simulation to 20π, while restricting the displayed plot to one oscillation.